

UprightBias

Rotated inputs degrades feed-forward reconstruction



VGGT-Ω

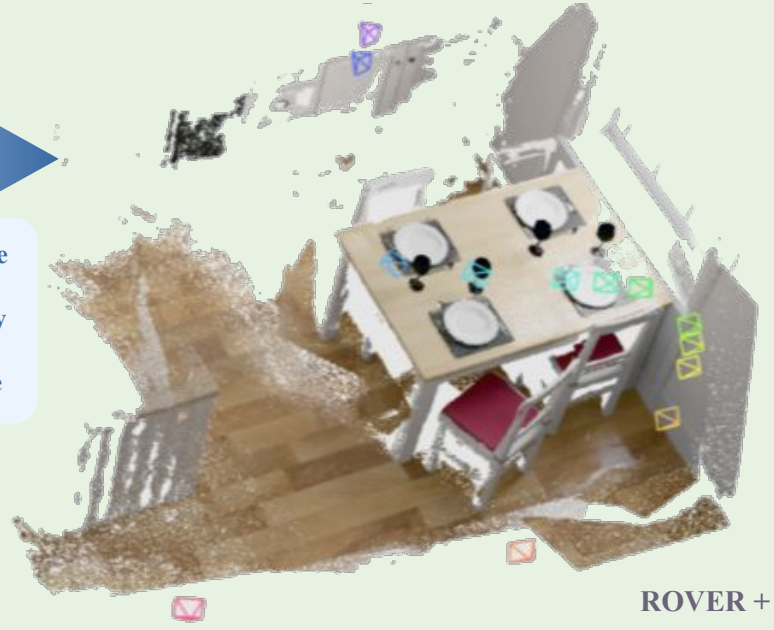
❌ Geometry collapse & inaccurate camera poses

ROVER

- Inference time
- Plug-and-Play
- Training-Free

With ROVER

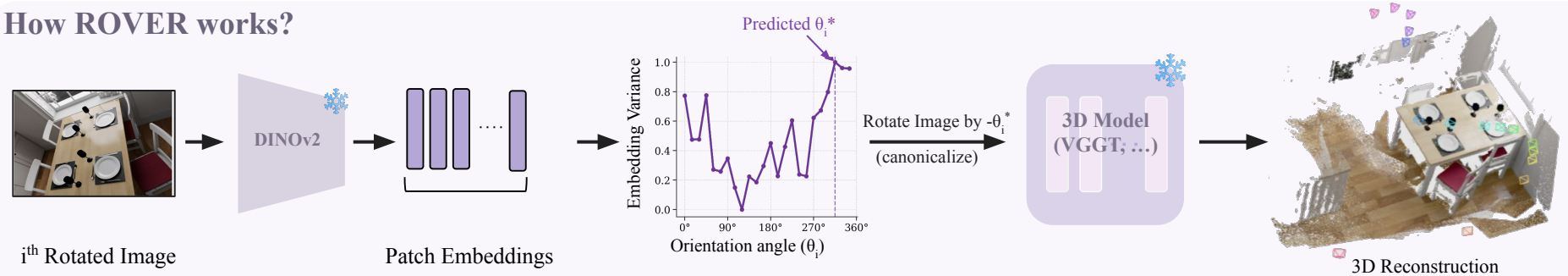
Accurate 3D reconstruction from the rotated inputs



ROVER + VGGT-Ω

✅ Geometry recovered & coherent camera poses

How ROVER works?



i^{th} Rotated Image

Patch Embeddings

3D Reconstruction